Replication package for:
Revealing Stereotypes: Evidence from Immigrants in Schools

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README

Overview

This README provides guidelines to replicate the analysis from "Revealing Stereotypes: Evidence from Immigrants in Schools" using Stata. It contains instructions for how to access the data and detailed guidance for replicating the results. One master do-file runs all of the code to analyse the data and produce the figures and tables in the main paper and Online Appendix. The estimated time to run this code is approximately 5 minutes.

Data Availability and Provenance Statements

The data used are from different sources:

1. **Teachers' Survey (Field Experiment):** the data was collected by the authors. The questionnaires are available in Appendix B.2 and the experimental instruction for the IAT is available in B.1.
    - Data.Name: AlesinaCarlanaLaFerraraPinotti_fieldexperiment_teachers.dta
    - We collected information on the name and surname of teachers, contact information, and school information. We used this information to assign a teacher ID and the identifiable elements were withheld.
2. **Teachers' Survey (Online Experiment):** the data was collected by the authors. The questionnaires are available in Appendix B.2 and the experimental instruction for the IAT is available in B.1.
    - Data.Name: AlesinaCarlanaLaFerraraPinotti_onlineexperiment.dta
    - We collected information on the name and surname of teachers, contact information, and school information. We used this information to assign a teacher ID and the identifiable elements were withheld.
3. **Data from the Italian Ministry of Education and INVALSI for the students in the field experiment:** the individual student data are confidential, but may be obtained with a Data Use Agreement ("Protocollo di Collaborazione") between universities, the Italian Ministry of Education and INVALSI. Researchers interested in accessing the data can contact Ufficio Statistica of the Ministry of Education (ufficio.statistico@miur.it) and INVALSI (uff.statistico@invalsi.it).
    - Data.Name: AlesinaCarlanaLaFerraraPinotti_fieldexperiment.dta

4. **INVALSI data in math for the universe of grade 8 students from 2011-12 to 2016-17:** the data can be obtained by requesting it to INVALSI following the directions provided here for special data requests: https://serviziostatistico.invalsi.it/en/richieste-dati-ad-hoc/
    - Data.Name: Invalsi_math_1112_1617.dta

Statement about Rights

We certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Summary of Availability

Data are publicly available, with the exception of student-level data from the field experiment in compliance with GDPR and the protocol of collaboration between the research team, INVALSI, and the Ministry of Education.

Details on each Data Source

In this section, we list the sources used to construct the main databases that were used in the analysis. For each one, we provide a description, the source, the folder location in the replication repository, and the analysis that requires the data. The names of the files containing data from each source are listed in Table A

Dataset list

The descriptions of all of the variables included in the analysis are detailed in file Codebook.xslx.

AlesinaCarlanaLaFerraraPinotti fieldexperiment

- **Description**: Contains data collected on students in the field experiment, including some sociodemographic characteristics.
- **Source**: Ministry of Education and INVALSI
- **Format**: The dataset is in `.dta` format.
- **Location in repository**: /Data/final/...

AlesinaCarlanaLaFerraraPinotti fieldexperiment teachers

- **Description**: Contains data collected on teachers in the field experiment, including some sociodemographic characteristics.
- **Source**: data collection from authors.
- **Format**: The dataset is in `.dta` format.
- **Location in repository**: /Data/final/...

AlesinaCarlanaLaFerraraPinotti onlineexperiment

- **Description**: Contains all data collected for the online experiment, including the baseline survey data and that collected after treatment for both randomized groups.
- **Source**: data collection from authors.
- **Format**: The dataset is in `.dta` format.

- **Location in repository**: /Data/final/...

Invalsi math 1112 1617

- **Description**: INVALSI data on student standardised test scores and sociodemographic characteristics
- **Source**: INVALSI
- **Format**: The dataset is in `.dta` format.
- **Location in repository**: /Data/other/...

Computational requirements

Software Requirements

Stata (code was last run with version 17, with all packages updated on 20/01/2024). The following packages are required to run the programs:

- `_gwmean`
- `ivreg2`
- `ranktest`
- `binscatter`
- `ritest`
- `estout`

They are all installed by switching the option `stata_programs = 1` in line 55 of the master do-file "`0_master_Revealing_Stereotypes.do`".

Controlled Randomness

We rely on a random number generator for the permutation test in Figure A.6 of the Appendix.

A seed is set in line 45 of the master dofile "`0_master_Revealing_Stereotypes.do`"

Memory and Runtime Requirements

The code was last run on a 4.2 GHz Quad-Core Intel Core i7 with MacOS version 13.6.6, with memory size 32 GB 2400 MHz DDR4. The code runs for less than 15 minutes.

Summary The approximate time needed to reproduce all of the analyses on a standard 2023 computer is 5 minutes.

Details The code was last run on a **14-core laptop with a 2.4 GHz Intel Core i7 processor and Microsoft Windows version 10.0**.

Description of Replication Folder

The replication folder is organised as follows:

- `Code`: this folder contains the scripts that run the code to replicate the analysis, tables, and figures in the paper and in the online appendix. One single master do-file runs all the necessary scripts.

- `Data` This folder provides the input datasets that are required to produce the analysis. This folder contains three subfolders:
  - `final`
  - `other`
  - `temp`
- `Output`: this folder contains all the figures (in PDF format) and the input (as .tex files) to produce the tables that are included in the paper. The code will store the results in the following subfolders.
  - `main`
    - `tables`
    - `figures`
  - `appendix`
    - `tables`
    - `figures`

Description of code

In this section, we describe the programs that are included in the Code subfolder of the replication folder.

1. `0_master_Revealing_Stereotypes.do` Runs in the corresponding order the other two Stata do-files below. It includes a section that installs all of the required commands and programs required to run the code, which can be switched off.
2. `1_EBE.do`. This code generates empirical Bayes estimates of teacher bias, and saves the resulting dataset in `temp` to be used later.
3. `2_tables_and_figures.do` This code produces the tables and figures in the main text and the appendix.

Instructions to Replicators

Download all of the data and code files in folders with names mimicking the ones in the repository.

- Change the main directory path in lines 51-52 of the Code/0_master_Revealing_Stereotypes.do do-file.
- Run the Code/0_master_Revealing_Stereotypes.do

Details

- Please note that the above instructions should be followed in order to reproduce the results correctly.
- If running independent parts of the code, it might be needed to run all the programs preceding any particular one.
- The construction of Figures 1 and A.1 and Table A.1 was not scripted on Stata.
  - Figure 1 is the timeline created by the authors.
  - Appendix Table A.1 contains data publicly available from the Italian Ministry of Education.

List of tables and programs

The provided code reproduces selected tables and figures in the paper, as explained and justified below. Table B lists all the Figures and Tables that are included in the Output folder of the replication folder.

Table A

Data.Name	Data.Files	Location	Provided	Citation
"Field experiment"	AlesinaCarlanaLaFerraraPinotti_fieldexperiment.dta	Data/final	TRUE	INVALSI and Italian Ministry of Education
"Field Experiment: Teachers"	AlesinaCarlanaLaFerraraPinotti_fieldexperiment_teachers	Data/final	TRUE	(Alesina, Carlana, La Ferrara, Pinotti, 2024)
"Online Experiment"	AlesinaCarlanaLaFerraraPinotti_onlineexperiment.dta	Data/final	TRUE	(Alesina, Carlana, La Ferrara, Pinotti, 2024)
"Invalsi Data"	Invalsi_math_1112_1617	Data/other	TRUE	INVALSI

Table B

Figure/ Table	**Program**	**Line Num.**	**Output file**	**Note**
Figure 1	Constructed by hand	N/A		Timeline of Data Collection for Field Exp.
Figure 2	2_tables_and_figures.do	82-132	F2dscore_sub_both.pdf	Distribution of race IAT score across teachers
Figure 3	2_tables_and_figures.do	135-176	F3gradesblindnoblind.pdf	Distribution of grades
Figure 4	2_tables_and_figures.do	178-212	F4graph_bias.pdf	Teacher-assigned vs. blind standardized grades
Figure 5	2_tables_and_figures.do	216-238	F5Figure_distribution_immigrant2.pdf	Correlation between Teacher Assigned Grades and IAT for Immigrants and Natives
Figure 6	2_tables_and_figures.do	240-307	F6graph_bias_by_IAT.pdf	Teacher-assigned vs. blind standardized test grades by Teacher IAT (High vs. Low)
Figure 7	2_tables_and_figures.do	311-427	F7results_exp_difference.pdf	Online Exp.: Impact of Revealing Stereotypes on grading of immigrant and native students
Table 1	2_tables_and_figures.do	432-626	T1balance_table_teacher_char.tex'	Balance table - Teacher characteristics
Table 2	2_tables_and_figures.do	629-1139	T2Correlation_iat_characteristics_v2.tex	Correlation between teacher characteristics and IAT
Table 3	2_tables_and_figures.do	1142-1293	T3bias_grading_subject_outcome'.tex'	Bias in Grading and Teachers' IAT scores
Table 4	2_tables_and_figures.do	1297-1850	T4teacher_revealbias.tex	Impact of revealing stereotypes on grades and on probability of failing students
Table 5	2_tables_and_figures.do	1853-1992	T5results_experiments.tex	Impact of revealing stereotypes by IAT
Table 6	2_tables_and_figures.do	1997-2135	T6results_experiment2_beliefupdating.tex	Impact of revealing stereotypes on grades (vs. general debiasing) in Online Exp.
Figure A.1	2_tables_and_figures.do	2143-2159	AF1correlation_graph.pdf	Teacher-assigned grades in Online Exp. vs. Original Grades
Figure A.2	2_tables_and_figures.do	2166-2264	AF2bias_IAT.pdf	Correlation between Bias in Grading and IAT
Figure A.3	2_tables_and_figures.do	2268-2316	AF3gradesblindnoblind_by_subject.pdf	Distribution of grades
Figure A.4	2_tables_and_figures.do	2383-2447	A4balance_table_intotreatment_students.tex	Teacher-assigned vs. blind standardized test grades by subject
Figure A.5	2_tables_and_figures.do	2320-2380	AF5grades_feedbacks.pdf	Field Exp.: The impact of Revealing Stereotypes to teachers on grading of immigrant and native students
Figure A.6	2_tables_and_figures.do	2450-2482	AF6Permutation_graph.pdf	Permutation test
Table A.1	Constructed by hand	N/A		Country of birth of immigrant students (2016-2017)

Figure/ Table	Program	Line Num.	Output file	Note
Table A.2	2_tables_and_figures.do	2597-2688	balance_table_sample_all_v2.tex	Balance between schools in the sample and out
Table A.3	2_tables_and_figures.do	2692-2689	A3balance_table_sample_exp1_v2.tex	Balance table - Teacher characteristics (Field Exp.)
Table A.4	2_tables_and_figures.do	2693-2798	A4balance_table_intotreatment_students.tex	Balance table - Student characteristics across schools included and not in the Exp.
Table A.5	2_tables_and_figures.do	2799-2861	A5Correlation_email_characteristics_v2.tex	Correlation between teachers characteristics and willingness to receive feedback
Table A.6	2_tables_and_figures.do	2865-2891	A6Correlation_iat_characteristics_add_v2.tex	Correlation between teachers characteristics and IAT (Field Exp.)
Table A.7	2_tables_and_figures.do	2895-2996	balance_table_students_v2.tex	Balance table - Student characteristics (Field Exp.)
Table A.8	2_tables_and_figures.do	2998-3121	A8bias_grading_subject_first_diff_std_grade.tex	Bias in Grading and Teachers' IAT scores
Table A.9	2_tables_and_figures.do	3123-3201	A9teacher_revealbias_transition.tex	Impact of revealing stereotypes on grades and on probability of failing students
Table A.10	2_tables_and_figures.do	3205-3367	A10teacher_revealbias_grade_IAT_wvs	Impact of revealing stereotypes on grades
Table A.11	2_tables_and_figures.do	3371-3487	A11results_experiment2_beliefupdating_2.tex	Impact of revealing stereotypes to teachers on grades (vs. general debiasing) in the Online Exp.
Table D.1	2_tables_and_figures.do	3490-3619	D1Abias_grading_subject_outcome'gs.tex; D1Bbias_grading_subjectoutcome'_gs.tex	Bias in Grading and Teachers' IAT scores
Table D.2	2_tables_and_figures.do	3626-3684	D2results_experiments_gs.tex	Impact of revealing stereotypes in the field and online Exp., by teacher IAT

References

Alesina Alberto, Michela Carlana, Eliana La Ferrara, and Paolo Pinotti, Revealing Stereotypes: Evidence from Immigrants in Schools, American Economic Review, Forthcoming.

Acknowledgements

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